The strict Courtney–Sarason lower bound for every self-adjoint Toeplitz truncation level

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Status. Candidate full solution of Question 1 in Courtney–Sarason [1]; likely valid subject to human review. Questions 2–4 of the source are not claimed solved.

1 Source question and notation

For $f \in L^\infty(\mathbb{T})$, let

$$A_{f,N} = \big(\widehat{f}(j-\ell)\big)_{j,\ell=0}^{N}$$

be the compression of multiplication by f to the polynomials $\mathcal{P}_N$. For a self-adjoint Toeplitz matrix A, let c_A be the least L^∞ norm of an inducing function with the prescribed Fourier coefficients, and set

$$c_N = \max_{0 \neq A = A^* \text{ Toeplitz}} \frac{c_A}{\|A\|}.$$

The source asks in Section 8, PDF page 13:

$$c_6 \geq 1.7505$$
$$c_7 \geq 1.7677.$$

8. Questions and Conjectures

We present some problems for future research suggested by the numerical results of Section 7, which tell us, in particular, that $c_N > \frac{\pi}{2}$ for $N = 2, 3, 4, 5, 6, 7$. (As noted in Section 7, for $N = 3$ the inequality can be verified by hand.) By the corollary to Proposition 7.1, we can conclude that $c_N > \frac{\pi}{2}$ for infinitely many N. We conjecture that the following question has an affirmative answer.

Question 1. *Is $c_N > \frac{\pi}{2}$ for all $N > 1$?*

Our estimates for c_1–c_7 suggest the following question.

Question 2. *Does c_N increase strictly as N increases?*

Again, we conjecture the answer is affirmative.

As noted in Section 1, the inequality $c_N \leq 2$ holds for all N, suggesting the next question.

Question 3. *What is $\sup\{c_N : N > 1\}$?*

Courtney–Sarason prove that the minimum-norm inducer is an alternating step function and, conversely, that every alternating step function of height one and order at most N is the minimum-norm inducer for its compression $A_{f,N}$ [1, Sections 4–5]. Consequently

$$c_N \geq \frac{1}{\|A_{f,N}\|} \tag{1}$$

for every such height-one step function f.

2 Proof intuition

The source dilates a symbol f to $f(z^k)$ only at the exact multiple level kN. The residue-class proof actually loses nothing at an arbitrary level M: each residue modulo k carries a Toeplitz block for f, and the largest block has level $\lfloor M/k \rfloor$. Thus a single strict level-two witness can be copied without changing its compression norm into every level $M \geq 4$ by taking $k = \lfloor M/2 \rfloor$. The only uncovered level is 3, where the source already supplies a rigorous strict witness.

3 The non-multiple dilation lemma

Lemma 1. *Let $f \in L^\infty(\mathbb{T})$, let M, k be positive integers with $k \leq M$, and put $g(e^{i\theta}) = f(e^{ik\theta})$. Then*

$$\|A_{g,M}\| = \|A_{f,\lfloor M/k \rfloor}\|. \tag{2}$$

More precisely, $A_{g,M}$ is unitarily equivalent to the direct sum

$$\bigoplus_{r=0}^{\min\{k-1,M\}} A_{f,\lfloor (M-r)/k \rfloor}. \tag{3}$$

Proof. The Fourier coefficient $\widehat{g}(n)$ is zero unless k divides n, and $\widehat{g}(kj) = \widehat{f}(j)$. For each residue $r \in \{0, \ldots, k-1\}$, let S_r be the span of the monomials

$$z^r, z^{r+k}, \ldots, z^{r+m_r k}, \qquad m_r = \left\lfloor \frac{M-r}{k} \right\rfloor,$$

omitting empty spaces. The spaces S_r are mutually orthogonal, span $\mathcal{P}_M$, and reduce $A_{g,M}$. Under the unitary identification $z^{r+jk} \leftrightarrow z^j$, the (p,q) entry of the restriction is

$$\widehat{g}(k(p-q)) = \widehat{f}(p-q),$$

so the block is A_{f,m_r}. This proves (3). The $r = 0$ block has level $m_0 = \lfloor M/k \rfloor$, while every other block is a principal compression of it. Therefore the norm of the direct sum is exactly the norm of the $r = 0$ block, proving (2). □

The exact-multiple case $M = kN$ is Proposition 7.1 of the source. The arbitrary M formulation gives an additional comparison.

Corollary 1. *For positive integers M, k with $k \leq M$,*

$$c_M \geq c_{\lfloor M/k \rfloor}.$$

Proof. Take an alternating step maximizer f at level $\lfloor M/k \rfloor$ and put $g(z) = f(z^k)$. Its order is at most $k\lfloor M/k \rfloor \leq M$, so (1) applies at level M. Lemma 1 preserves the relevant compression norm. □

4 A rigorous strict seed at level two

Let ψ be the even alternating step function of height one and order two which equals 1 at $1 \in \mathbb{T}$ and has discontinuities at the angles

$$\pm L, \quad \pi \pm R, \qquad L = \frac{\pi}{15}, \quad R = \frac{\pi}{3}.$$

Direct integration (the same formula used in Section 7 of the source) gives

$$a_0 := \widehat{\psi}(0) = -\frac{1}{5}, \tag{4}$$

$$a_1 := \widehat{\psi}(1) = \frac{2}{\pi}\left(\sin\frac{\pi}{15} - \frac{\sqrt{3}}{2}\right), \tag{5}$$

$$a_2 := \widehat{\psi}(2) = \frac{1}{\pi}\left(\frac{\sqrt{3}}{2} + \sin\frac{2\pi}{15}\right). \tag{6}$$

Thus

$$A_{\psi,2} = \begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_0 & a_1 \\ a_2 & a_1 & a_0 \end{pmatrix}.$$

Put $x = |a_1|$ and $y = a_2$. The antisymmetric eigenvector gives the eigenvalue $-1/5 - y$, and the two symmetric eigenvalues are

$$-\frac{1}{5} + \frac{y}{2} \ \pm \ \frac{1}{2}\sqrt{y^2 + 8x^2}. \tag{7}$$

For completeness, all numerical separation needed below follows from the rational bounds

$$3.14159 < \pi < 3.14160, \qquad 0.20791 < \sin(\pi/15) < 0.20792,$$
$$0.40673 < \sin(2\pi/15) < 0.40674, \qquad 0.86602 < \sqrt{3}/2 < 0.86603.$$

Machin's formula and alternating series prove the first bound; alternating Taylor bounds prove the two sine bounds; the last follows by squaring. They imply

$$x < 0.41898, \qquad 0.40512 < y < 0.40514.$$

Hence the antisymmetric eigenvalue has modulus below 0.60515, while

$$\left|-\frac{1}{5} + \frac{y}{2}\right| < 0.00257, \qquad \frac{1}{2}\sqrt{y^2 + 8x^2} < 0.62620.$$

By (7),

$$\|A_{\psi,2}\| < 0.62877 < \frac{2}{\pi}. \tag{8}$$

The accompanying verifier certifies each displayed decimal inequality by exact rational arithmetic. From (1) and (8), $c_2 > \pi/2$.

5 Full answer to Question 1

Theorem 1. *For every integer $N > 1$, the Courtney–Sarason mini-max constant satisfies*

$$c_N > \frac{\pi}{2}.$$

Proof. The case $N = 2$ is (8). For $N = 3$, Section 7 of the source constructs an alternating step function proving

$$c_3 \geq \frac{\pi\kappa}{2\sqrt{2}},$$

where κ is the largest root of $1 - 3x - 3x^2 + 3x^3$. Since this polynomial is negative at $3/2$, $\kappa > 3/2$; consequently $c_3 > 3\pi/(4\sqrt{2}) > \pi/2$.

Now let $N \geq 4$ and set $k = \lfloor N/2 \rfloor$. Then

$$\left\lfloor \frac{N}{k} \right\rfloor = 2, \qquad 2k \leq N.$$

Equivalently, the dilated function $g(z) = \psi(z^k)$ is an alternating step function of order $2k \leq N$, and Lemma 1 gives

$$\|A_{g,N}\| = \|A_{\psi,2}\| < \frac{2}{\pi}.$$

Applying (1) to g yields $c_N > \pi/2$. ∎

6 Verification, novelty, and limitations

The standard-library verifier proves the rational interval certificate, then checks the residue block levels for every $4 \le N \le 1000$. The finite sweep is only a sanity check; Lemma 1 proves the statement for all N.

A bounded run-index and arXiv/web search on 13 August 2026 used the exact title, authors, c_N, "self-adjoint Toeplitz $\pi/2$," "Courtney–Sarason," and the exact wording of Questions 1–2. It recovered the source and later papers citing it for other Toeplitz/Blaschke questions, but no resolution or non-multiple dilation formula. The novelty assessment is therefore *apparently new within the bounded search*, not an exhaustive guarantee.

Theorem 1 completely answers Question 1. Corollary 1 gives new non-adjacent comparisons, but it does not imply $c_{N+1} > c_N$. Questions 2–4 remain open. Human review should prioritize the source-derived extremal reformulation and the arbitrary-level block decomposition.

References

[1] D. Courtney and D. Sarason, *A mini-max problem for self-adjoint Toeplitz matrices*, Math. Scand. **110** (2012), 82–98; doi:10.7146/math.scand.a-15198; arXiv:1009.2553.